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Clinical study

Comparison of efficiency between VNS and ANT-DBS therapy in drug-resistant epilepsy: A one year follow up study

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ABSTRACT

Vagus nerve stimulation (VNS) and anterior thalamic deep brain stimulation (ANT-DBS) have both been used for treatments of drug-resistant epilepsy (DRE). However, there is no comparative study on the effectiveness of two methods from one single center. 17 patients with DRE who underwent VNS therapy and 18 patients who underwent DBS were enrolled. A retrospective study was performed starting from baseline before operation extending to 12 months after operation. The seizure types, duration of epilepsy, age at implantation, failed numbers of antiepileptic drugs (AEDs) before operation, history of craniotomy, stimulation parameters and response rate were described. The analysis of liner regression on the age of onset, duration of epilepsy, numbers of AEDs, and the seizure reduction at 12 months after operation was applied. The mean seizure reduction in patients with DBS at 3, 6, 9 and 12 months after the operation was 57.22%, 61.61%, 63.94% and 65.28%, and that in cases with VNS was 36.06%, 39.94%, 45.24% and 48.35%, respectively. At 1 year after the operation, the patients with older operation age, focal seizures and older age of onset responded better to VNS; and those older operation age, focal generalized seizures, history of craniotomy and longer duration of disease responded better to DBS. The efficiency of ANT-DBS was higher than that of VNS at each follow up time point. Patients can choose the appropriate treatment according to the individual clinical characteristics.

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1. Introduction

Epilepsy is a common neurological disorder, which characterized by spontaneous and recurrent seizures[1]. There are about 70 million patients all around the world, and the treatment of epilepsy mainly includes drug treatment, resective surgery and neuromodulation[2]. Unfortunately, 30% of epilepsy patients are classified into the DRE who cannot control seizures after more than 2 years of treatment with $\geq$ 2 types of AEDs[3]. Millions of patients with DRE cannot undergo the resective surgery or still experience recurrent seizures despite surgery[4]. Fortunately, the neuromodulation is an adjuvant therapy for drug and surgical resection, which can significantly reduce seizures and improve cognitive function[5].

At present, VNS and ANT-DBS are the mainly used neuromodulation, and have been respectively approved by American Food and Drug Administration (FDA) for the treatment of patients with DRE in 1997 and 2018[6]. The previous studies have proved that both methods were beneficial for patients with poor seizure control who are not suitable for the craniotomy surgery[7]. A 5 year follow up study reported that the mean seizure reduction of patients with VNS was 63.6%, and 64.4% of them had a reduction of more than 50%[8]. The SANTE trail with 110 cases in 2015 indicated that the mean reduction rate of epilepsy was 69% in patients with ANT-DBS, and the response rate ($\geq$50%) was 68%[9].

However, there is no uniform standard for the selection of surgical treatment and prediction of therapeutic effect, and neurosurgeons can only choose the appropriate method according to personal experience and patients' requirements[5]. A scientific basis for rational selection between different neuromodulation therapies is in great request. In this study, we recruited epilepsy patients who underwent VNS or ANT-DBS from a single medical center in china, and reported the treatment outcome of two method. This study was approved by the ethics committee of Xuanwu Hospital, Capital Medical University, China.

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2. Methods

This study included 35 DRE patients with detailed postoperative 12 months follow-up data who underwent VNS or ANT-DBS therapy between June 2013, and July 2018. All patients underwent routine preoperative evaluation by the Comprehensive Epilepsy Center, Xuanwu Hospital, which includes several neurosurgeons, neurologists, electrophysiologists, radiologist, and other researchers. Considering the long-term (interictal and ictal) video-EEG, standard magnetic resonance imaging, positron emission tomography, magnetoencephalography and clinical manifestations, a resection operation was ruled out and an optimal VNS or ANT-DBS surgery plan was worked out. The selection between two treatments followed the following principles: (1) the recommendation of preoperative consultation was VNS or ANT-DBS; (2) the type of operation was voluntarily chosen by patients and guardians after detailed surgical risk notification.

Their electronic medical records and auxiliary examination results (electroencephalogram, magnetoencephalography, magnetic resonance imaging, and so on) were reviewed, and the following data were also collected: patient demographics, seizure characteristics, preimplant diagnostic tests, stimulation titration schedule, seizure frequency before and after implantation). The patients were followed up in one half day clinic every 1–3 month by the doctors who performed the operation, depending on the therapeutic effect.

For patients with VNS, the VNS device was implanted under general anesthesia by exposing the left vagus nerve located medial to the jugular vein. A coil electrode (Cyberonics, Houston, TX, USA) was wrapped around the vagus nerve and the lead was fixed using silicon anchors, as recommended by the manufacturer. An internal pulse generator (IPG, Cyberonics) was subcutaneously implanted into the left upper chest.

For patients with ANT-DBS, the leads (Medtronic, Framingham, MA, USA) were stereotactically implanted under general anesthesia using visual targeting based on 3.0-tesla magnetic resonance images acquired using a Leksell Coordinate Frame G (Elekta AB, Stockholm, Sweden) fixed on the patient's cranium; T2-weighted and gadolinium-enhanced T1-weighted images were acquired at 1-mm slice thickness. The DBS leads were targeted at the bilateral ANT in all patients. Finally, an implantable IPG (Activa® PC [37601] or Activa® RC [37612], Medtronic) was connected to the electrode and implanted in the left subclavian area.

The stimulation titration schedule of VNS was modified according to therapeutic effects: (1) the primary stimulus levels were 0.25 mA (current intensity), 30 Hz (signal frequency), 250 μs (pulse width), 30 s/5 min (on time/off time) at 2 weeks post-implantation; (2) the output current intensity was increased 0.5 mA every 1 month interval; (3) when the current intensity reached 1.5 mA, patients were followed-up every 3 months and the parameters were modified according to therapeutic effects (output current was increased by 0.25 mA at every titration).

The stimulation titration schedule of DBS was modified individually according to therapeutic effects to reach the best clinical outcome. The optimum stimulation parameters for a patient were defined as the highest parameter that he/she can tolerate and that provided maximum seizure reduction. We choose the recommended parameters: ≥100 Hz, 1–10 V, and the electrode contacts and duty cycles were adjusted smoothly within the patients' tolerance[10].

Seizure frequencies and types before VNS and DBS were retrospectively evaluated from the patient records of all patients. The seizure reduction was defined using the equation: (seizure/month on VNS − baseline seizure / month) / (baseline seizure / month) × 100%. Baseline seizure frequency was defined as the average of the monthly seizure counts over a 6-month period before VNS or DBS implantation. After the implantation, the seizure frequency was measured by interview at each outpatient visit for at least 12 months and summarized with a 3-month interval. In addition, we investigated which seizure types were well controlled after the surgery for each subject.

The analysis of liner regression on the age of onset, duration of epilepsy, numbers of AEDs, and the seizure reduction at 12 months after operation was applied in the GraphPad Prism 8.0.1 (https://www.graphpad.com/). A significant P-value was established at <0.05.

3. Results

3.1. Population demographics

17 epilepsy patients (20.24 ± 11.40 years old, range 5–41 years old) with VNS and 18 patients with ANT-DBS (28.94 ± 12.00 years old, range 12–52 years old) were enrolled in the study. There were 8 patients with focal seizures, 9 patients with generalized seizures in VNS group, and 13 subjects with focal seizures and 5 subjects with generalized seizures in DBS group. The clinical features and treatment responses of patients are summarized in Table 1 and Table 2.

3.2. The outcome of VNS

The mean seizure reduction at 3, 6, 9 and 12 months after the operation was as follows: 36.06%, 39.94%, 45.24% and 48.35% in all 17 patients with VNS; 28.00%, 32.50%, 38.30% and 42.20% in 10 children patients (<18 years old), which was 47.57%, 50.57%, 55.14% and 57.14% in 7 adults patients (≥18 years old); 28.75%, 39.25%, 46.13% and 59.00% in 8 patients with focal seizures, and 42.56%, 40.56%, 44.44% and 38.89% in 9 cases with generalized seizures; 29.00%, 32.80%, 41.20% and 46.80% in 5 patients with history of craniotomy, while 39.00%, 42.92%, 46.92% and 49.00% in 12 cases without history. At the last follow-up (12 months), 58.82% (N = 10) reported ≥50% reduction in seizures, 17.65% (N = 3) were seizure free, while 41.18% (N = 7) were non-responders; patients may response better to those with older age of onset ($P < 0.05$); there was no significant correlation between the duration of disease or numbers of AEDs and the seizure reduction ($P > 0.05$) (Fig. 1).

The mean VNS settings at the end of follow-up were the following: output current, 1.82 ± 0.26 mA (1.5 mA for 4 patients, 1.75 mA for 6 patients, 2.0 mA for 6 patients, 2.5 mA for 1 patient); signal frequency, 30 Hz (30 Hz for all patients); pulse width, 367.65 ± 128.62 μs (250 μs for 9 patients, 500 μs for 8 patients); on time, 30 s (30 s for all patients); off time, 4.65 ± 0.79 min (3 min for 3 patients, 5 min for 14 patients) (Fig. 2).

3.3. The outcome of ANT-DBS

The mean seizure reduction at 3, 6, 9 and 12 months after the operation was as follows: 57.22%, 61.61%, 63.94% and 65.28% in all 18 patients with DBS; 52.33%, 56.33%, 56.33% and 52.33% in 3 patients < 18 years old, and 58.20%, 62.67%, 65.47% and 67.87% in 15 patients ≥ 18 years old; 63.92%, 64.31%, 69.69% and 71.15% in 13 patients with focal seizures, and 39.80%, 54.60%, 49.00% and 50.00% in 5 cases with generalized seizures; 64.60%, 64.20%, 73.60% and 81.20% in 5 patients with history of craniotomy, while 54.38%, 60.62%, 60.23% and 59.15% in 13 cases without history. At the last follow-up (12 months), 72.22% (N = 13) of patients were responders (self-reported ≥ 50% reduction in seizures), 22.22% (N = 4) were seizure free, while 27.78% (N = 5) were non-responders; patients may response better to ones with longer duration of epilepsy ($P < 0.05$); there was no significant correlation

Table 1
Demographics and clinical details of patients with implanted VNS.

No.	Sex	Age of onset	Duration of epilepsy	Operation age	Seizure types	AEDs (no.)	History of craniotomy	Seizure reduction (ao)				AEDs at the surgery	AEDs at 12 m ao
								3 m	6 m	9 m	12 m		
1	F	1y	9y	10y	F	5	No	0%	0%	50%	63%	Le	Le
2	F	3y	2y	5y	F	6	No	0%	50%	50%	75%	To + Le + Va	To + Le + Va
3	F	12y	13y	25y	F	2	EEF (R, F)	0%	0%	0%	0%	Cl + La + Ph + Va	Cl + La + Ph + Va
4	M	3d	8y	8y	G	4	No	0%	0%	0%	0%	Ni + To + Va	Ni + To + Va
5	M	2y	7y	9y	G	7	No	100%	75%	50%	0%	Cl + La + To + Va	Cl + La + To + Va
6	M	2y	7y	9y	G	4	No	80%	80%	85%	50%	Cl + L + Va	Cl + L + Va
7	M	6y	31y	37y	G	5	ATL + H + C (R)	65%	50%	80%	80%	La + Le	La + Le
8	M	8y	33y	41y	G	2	ATL + H + C (R)	0%	0%	0%	0%	Ox + Ph	Ox + Ph
9	M	10y	2y	12y	G	5	No	0%	0%	0%	25%	Le + Ni + To + Va	Le + Ni + To + Va
10	M	10y	4y	14y	F	3	No	0%	0%	0%	0%	Cl + Va + Le	Cl + Va + Le
11	M	10y	4y	14y	G	6	No	50%	50%	75%	75%	La + Le + To	La + Le + To
12	M	11y	6y	17y	F	3	EEF (R, F)	0%	20%	30%	54%	La + Le + Va	La + Le + Va
13	M	11y	14y	25y	F	4	No	50%	50%	43%	80%	Ca + La + Le + Va	Ca + La + Le + Va
14	M	12y	13y	25y	G	3	No	0%	10%	10%	20%	Ca + Va	Ca + Va
15	M	18y	11y	29y	G	2	No	88%	100%	100%	100%	Ca + Va	Ca + Va
16	M	23y	4y	27y	F	3	HE (L, F + T)	80%	94%	96%	100%	Ox + Le + Va	Ox + Le + Va
17	M	27y	10y	37y	F	2	No	100%	100%	100%	100%	Ox	Ox

Abbreviations: ao, after operation; ATL, anterior temporal lobectomy; C, Callosotomy; EEF, excision of epileptogenic focus; F in History of craniotomy, Frontal; F in Sex, female; F in Seizure types, focal seizure; G, generalized seizure; H, Hippocampectomy; HE, hematoma evacuation for traumatic intracranial hemorrhage; M, male; T in History of craniotomy, Temporal. y in Age of onset and Operation age, years old; y in Duration of epilepsy, years. Ca in AEDs, Carbamazepine; Cl in AEDs, Clonazepam; La in AEDs, Lamotrigine; Le in AEDs, Levetiracetam; Ni in AEDs, Nitrazepam; Ox in AEDs, Oxcarbazepine; Ph in AEDs, Phenobarbital; To in AEDs, Topiramate; Va in AEDs, Valproate.

Table 2
Demographics and clinical details of patients with implanted ANT-DBS.

No.	Sex	Age of onset	Duration of epilepsy	Operation age	Seizure types	AEDs (no.)	History of craniotomy	Seizure reduction (ao)				AEDs at the surgery	AEDs at 12 m ao
								3 m	6 m	9 m	12 m		
1	F	3y	15y	18y	G	4	No	62%	78%	74%	75%	La + Le + To	La + Le + To
2	F	7y	15y	22y	F	1	No	100%	100%	100%	100%	Ox	Ox
3	F	10y	8y	18y	F	4	No	31%	6%	6%	19%	Le + Ox	La + Le
4	F	10y	17y	27y	F	2	A (R, Am)	27%	36%	45%	36%	Ca + Ph	Ca + Ph
5	F	12y	10y	22y	G	3	No	0%	29%	29%	43%	La + Ox + Va	La + Ox + Va
6	F	23y	3y	26y	G	5	No	80%	76%	52%	68%	Cl + Le + Ox	Cl + Le + Ox
7	F	24y	25y	49y	F	3	No	100%	100%	100%	100%	Ca + To	Ca + To
8	F	28y	17y	45y	F	5	DC (R, P)	50%	50%	75%	100%	Ca	Ca
9	M	5y	7y	12y	G	3	No	57%	40%	40%	14%	Ca + Cl + La	Ca + Ph + Va
10	M	6y	12y	18y	F	2	No	13%	30%	46%	71%	Va	Le + Va
11	M	9y	18y	27y	F	2	ATL (L)	95%	90%	84%	92%	Ca + Ph	Ca + Ph
12	M	12y	10y	22y	F	2	No	50%	100%	75%	50%	Ox + Va	Ox + Va
13	M	12y	16y	28y	F	6	DC (L, P)	84%	89%	89%	89%	Le + Ni + Ox + Ph	Le + Ni + Ox + Ph
14	M	12y	17y	29y	F	5	DC (R, F)	67%	56%	75%	89%	Ca + Cl + To	Ca + To
15	M	14y	36y	50y	F	3	No	85%	90%	90%	100%	Le	Le
16	M	15y	37y	52y	G	3	No	0%	50%	50%	50%	Le + Ox + Va	Le + Ox + Va
17	M	17y	10y	27y	F	6	No	29%	14%	71%	29%	Ca + La + Ph	Ca + La + Ph
18	M	25y	4y	29y	F	4	No	100%	75%	50%	50%	Ca + Le + Va	Ca + Le + Va

Abbreviations: A, ablation; Am, amygdala; ao, after operation; ATL, anterior temporal lobectomy; DC, decompressive craniectomy; F in Sex, female; F in Seizure types, focal seizure; F in History of craniotomy, Frontal; G, generalized seizure; L, left; M, male; P, Parietal; R, right. y in Age of onset and Operation age, years old; y in Duration of epilepsy, years. Ca in AEDs, Carbamazepine; Cl in AEDs, Clonazepam; La in AEDs, Lamotrigine; Le in AEDs, Levetiracetam; Ni in AEDs, Nitrazepam; Ox in AEDs, Oxcarbazepine; Ph in AEDs, Phenobarbital; To in AEDs, To in AEDs, Topiramate; Va in AEDs, Valproate.

between the age of onset or numbers of AEDs and seizure reduction ($P > 0.05$) (Fig. 3).

The mean DBS settings at the end of follow-up were the following: left voltage, 2.17 ± 0.35 V (1.5 V for 1 patient, 2.0 V for 11 patients, 2.5 V for 4 patients, 2.6 V for 1 patient, 3.0 V for 1 patient); right voltage, 2.2 ± 0.31 V (2.0 V for 12 patients, 2.5 V for 4 patients, 2.6 V for 1 patient, 3.0 V for 1 patient); frequency, 128.61 ± 18.05 Hz (60 Hz for 1 patient, 130 Hz for 14 patients, 145 Hz for 3 patients); pulse width, 90.56 ± 2.36 μs (90 μs for 17 patients, 100 μs for 1 patient); on time, 1 min (1 min for all patients); off time 4.72 ± 0.46 min (4 min for 5 patients, 5 min for 13 patients).

4. Discussion

The International League Against Epilepsy (ILAE) has defined the DRE as a failure of adequate trials of at least 2 antiepileptic drugs (AEDs) that are appropriately chosen, used, and tolerated [11]. The VNS and ANT-DBS are widely used in the treatment of patients with DRE, and proved to be safe and effective for various types of seizures by the previous studies [6,12]. There are currently no definitive data to suggest the suitable patients to any of the neurostimulation modalities, and it is impossible to choose optimum treatments for patients individually. Little attention has been

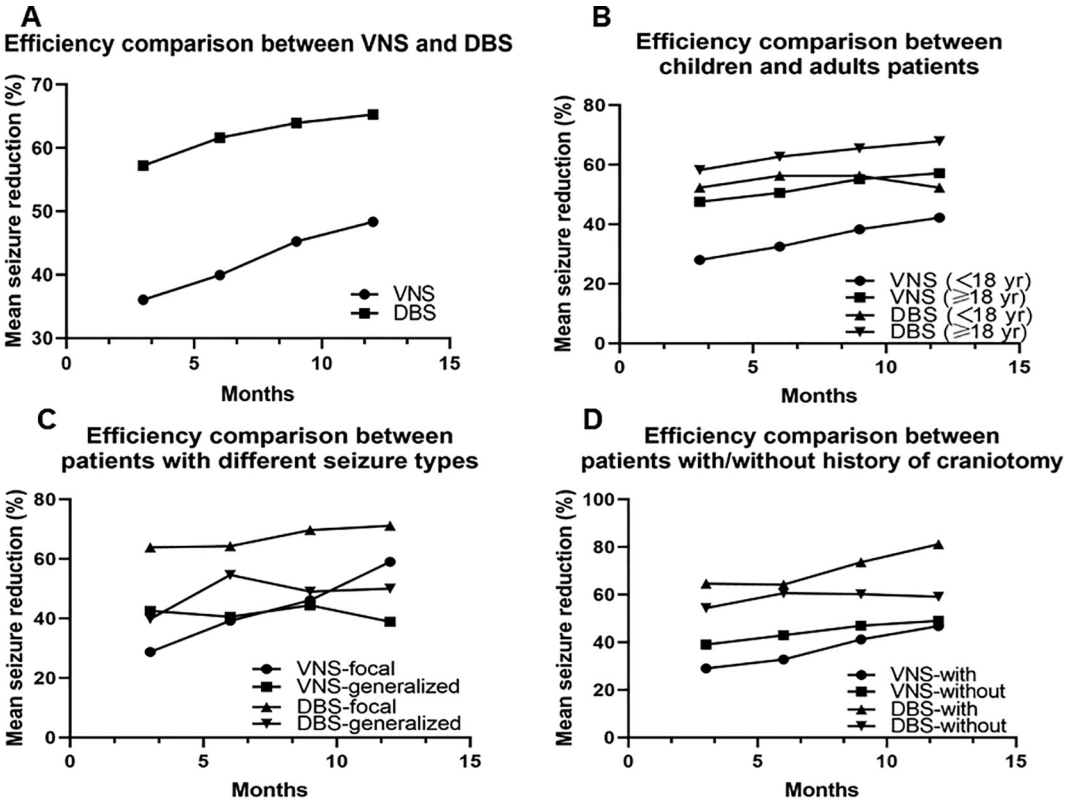

Fig. 1. The efficiency comparison between VNS and ANT-DBS group. (A) Efficiency comparison between VNS and ANT-DBS group. The mean seizure reduction in patients with VNS (N = 17) at 3, 6, 9 and 12 months after operation was 36.06%, 39.94%, 45.24% and 48.35%, and that in cases with DBS (N = 18) was 57.22%, 61.61%, 63.94% and 65.28%, respectively. (B) Efficiency comparison between children and adults patients. The mean seizure reduction in children patients with VNS (<18 years old, N = 10) at 3, 6, 9 and 12 months after operation was 28.00%, 32.50%, 38.30% and 42.20%, and 47.57%, 50.57%, 55.14% and 57.14% in adults patients (≥18 years old, N = 7). The mean seizure reduction in children patients with ANT-DBS (<18 years old, N = 3) at 3, 6, 9 and 12 months after operation was 52.33%, 56.33%, 56.33% and 52.33%, and 58.20%, 62.67%, 65.47% and 67.87% in adults patients (≥18 years old, N = 15). (C) Efficiency comparison between patients with different seizure types. In the VNS group, the mean seizure reduction in patients with focal seizures (N = 8) at 3, 6, 9 and 12 months after operation was 28.75%, 39.25%, 46.13% and 59.00%, and 42.56%, 40.56%, 44.44% and 38.89% in cases with generalized seizures (N = 9). In the ANT-DBS group, the mean seizure reduction in patients with focal seizures (N = 13) at 3, 6, 9 and 12 months after operation was 63.92%, 64.31%, 69.69% and 71.15%, and 39.80%, 54.60%, 49.00% and 50.00% in cases with generalized seizures (N = 5). (D) Efficiency comparison between patients with/without history of craniotomy. In the VNS group, the mean seizure reduction in patients with the previous craniotomy (N = 5) was 29.00%, 32.80%, 41.20% and 46.80%, while that in in cases without history (N = 12) was 39.00%, 42.92%, 46.92% and 49.00%. In the ANT-DBS group, the mean seizure reduction in patients with the previous craniotomy (N = 5) was 64.60%, 64.20%, 73.60% and 81.20%, while that in cases without history (N = 13) was 54.38%, 60.62%, 60.23% and 59.15%.

paid on the comparison of two neuromodulation techniques (VNS and ANT-DBS) in a single medical center.

VNS was approved by FDA for the treatment of ≥ 4 years old patients with DRE [13]. The seizure control improved over time with median seizure reduction was 25.0%, 40.9% and 53.3%, and responder rates of 38.9%, 46.8% and 55.8% at 3, 6, 12 months of VNS therapy, respectively [14]. Some evidence suggested that generalized seizures, a shorter history of seizures, no prior intracranial epilepsy surgery, younger operation age were predictors of favorable response to VNS [15–18]. While, other studies found that those with focal seizure may achieve better prognosis, and there was no correlations between the operation age, seizure types, history of craniotomy, and seizure reduction [19–22]. In addition, it was reported that the clinical data such as age, sex, seizure type, epilepsy syndrome, etiology, and seizure frequency could not predict the effect of VNS [23].

The bilateral stimulation of ANT using the Medtronic DBS System approved by FDA in 2018 as an adjunctive therapy for reducing the frequency of seizures in individuals 18 years of age or older in cases of medically refractory epilepsy [24]. It was once regarded only suitable for the ones with focal seizures [25]. The patients with younger operation age and shorter duration of epilepsy may be the responders [10]. While, the analyses stratifying for patients' demographics, epilepsy type or epilepsy cause, were inconclusive due to the short application time. The parameters of ANT stimulation still remain empirical, and the recommended parameters in the SANTE trail (frequency 145 Hz, 90 us pulse width, at voltages of 5 V, a cycle time of 1 min on and 5 min off) will achieve an mean seizure reduction rate of 76% in 5 years after operation [9,26]. The previous finding of our team was that the surgical effect is dependent on the frequency of stimulation, and lower frequency (<10 Hz) may increase the risk of seizures, while higher frequency (>100 Hz) could reduce the seizures [27]. It can also be considered as a further treatment option for refractory epilepsy patients even if they have previously experienced VNS failure [4].

The data from different centers may lead to inconsistent conclusions, and it is worth to compare the different methods from a single center to eliminate the bias of patients' selection criteria and surgical techniques. In this study, we recruited patients with VNS or ANT-DBS who were evaluated and operated by the same team of experts. The results showed that the mean seizure reduction of VNS and ANT-DBS at 12 months after operation was 48.35% and 65.28%, and both methods have accumulative effects over time. At 1 year after the operation, the adult patients with focal seizures responded better to both VNS and ANT-DBS. The older age of onset in VNS, and history of craniotomy and longer duration of epilepsy in ANT-DBS may indicate a better prognosis. The most tolerated parameters, at the end of follow-up, was stimulation current ranged between 1.75-2.0 mA in patients with VNS, and the voltage between 2.0 and 2.5 V in those with ANT-DBS.

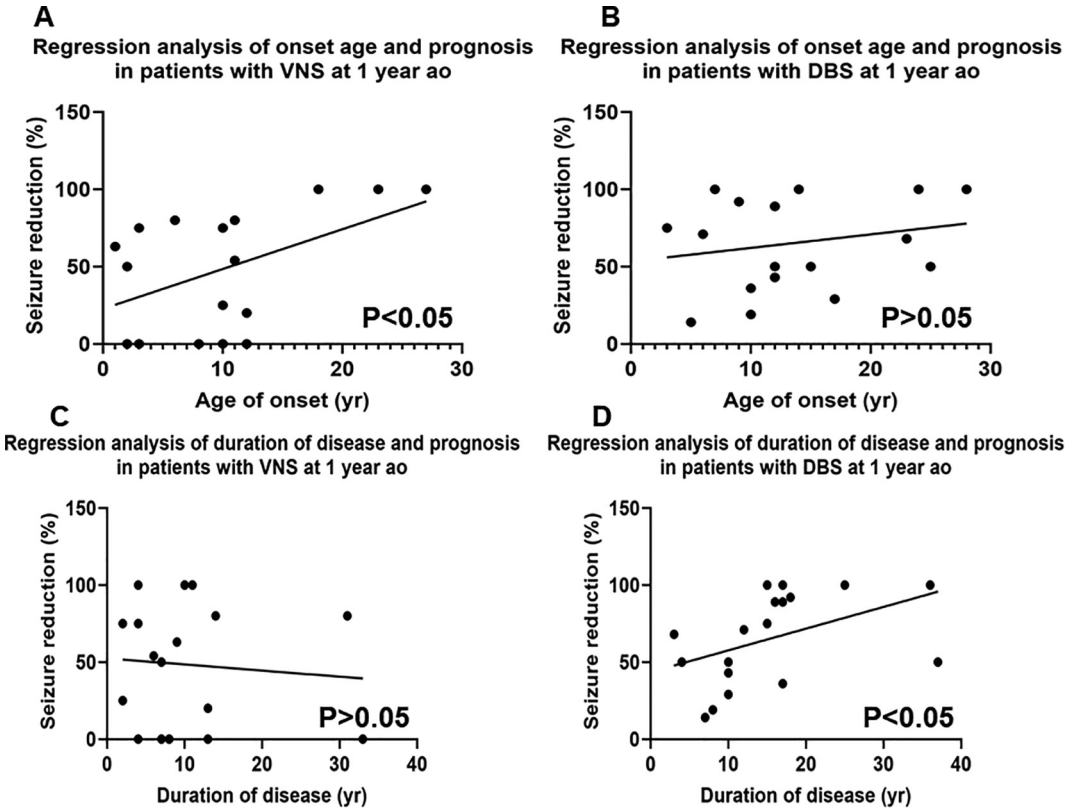

Fig. 2. The regression analysis of onset age, duration of disease and prognosis in patients at 12 months after operation. (A) The regression analysis of onset age and prognosis in patients with VNS at 12 months after operation. Epilepsy patients may response better to those with older age of onset ($P < 0.05$). (B) The regression analysis of onset age and prognosis in patients with ANT-DBS at 12 months after operation. There was no significant correlation between the age of onset and seizure reduction at 1 year after the operation ($P > 0.05$). (C) The regression analysis of epilepsy duration and prognosis in patients with VNS at 12 months after operation. There was no significant correlation between the duration of epilepsy and seizure reduction at 1 year after the operation ($P > 0.05$). (D) The regression analysis of epilepsy duration and prognosis in patients with ANT-DBS at 12 months after operation. Epilepsy patients may response better to those with longer duration of epilepsy ($P < 0.05$). Abbreviation: ao, after operation.

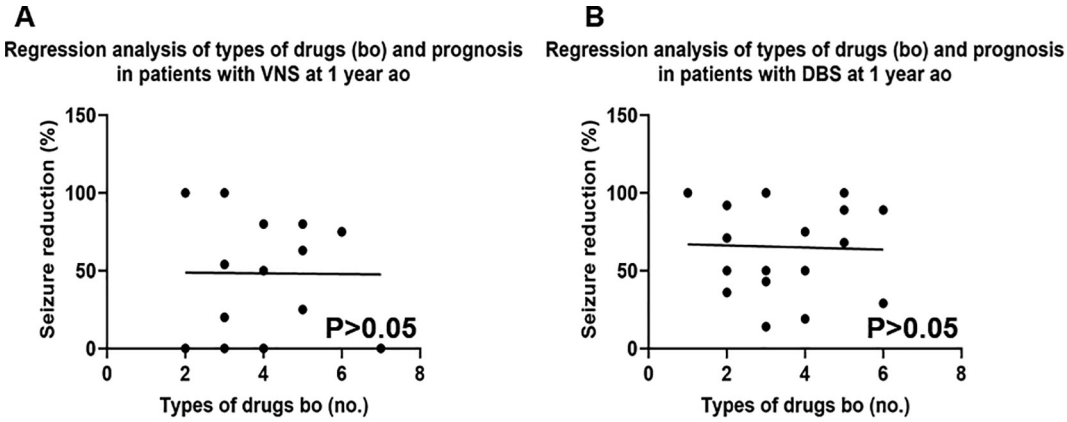

Fig. 3. The regression analysis of drug numbers and prognosis in patients at 12 months after operation. (A) The regression analysis of drug numbers and prognosis in patients with VNS at 12 months after operation. There was no significant correlation between the numbers of drugs and seizure reduction at 1 year after the operation ($P > 0.05$). (B) The regression analysis of drug numbers and prognosis in patients with ANT-DBS at 12 months after operation. There was no significant correlation between the numbers of drugs and seizure reduction at 1 year after the operation ($P > 0.05$). Abbreviation: bo, before operation.

Our study confirmed that VNS and ANT-DBS were valid options in the treatment for patients with DRE; the total seizure reduction of ANT-DBS was better than VNS in patients with different clinical characteristic; VNS was beneficial to patients with older age of onset, older age at operation and focal seizures; ANT-DBS was suitable for those with older age at operation, focal seizures, history of craniotomy, and longer duration of epilepsy; there seemed no correlation between the numbers of preoperative AEDs and outcome. The reason of lower mean seizure reduction in comparison with previous studies may be that the more strictly preoperative evaluation was performed in our department.

With the advancement of technology in the field of diagnosis and treatment of epilepsy, the challenge ahead of us in clinical practice is to integrate the different neuromodulation methods and effectively control the repeated seizures. The selection of patients may be important when evaluating different implantation,

and this one year follow up research provided a possible selection of treatment based on clinical data of patients before operation.

Although the parameter schedule and anti-epileptic mechanism of two operations seem dissimilar, the continuous stimulation during follow-up period approved the safety of operations. It has been reported that the most common complications were hoarseness, cough, pain, and dyspnea in VNS, which were implant site pain, infection, and lead misplacement in DBS [28]. The stimulation titration schedule of VNS and DBS used in this study was modified according to therapeutic effects and adjusted smoothly within the patients' tolerance. No serious complications or side effects were reported until the last follow-up time point.

The duration of stimulation is probably a critical factor affecting the outcome of surgery. There's a lot of clinical evidence showed that the seizure outcome of the VNS can further improve over the next years [29,30]. The limitation of this study is a relatively small number of cases and short observation period, and we did not use statistical methods for analysis. But the descriptive results did provide us with some new understanding of neuromodulation for epilepsy.

5. Conclusions

This is the first study in which the therapeutic response to chronic VNS and ATN-DBS in patients with DRE were assessed. The results showed the feasibility of treatment selection according to preoperative clinical data. Further studies with larger numbers of subjects, different epileptic syndromes and longer follow-up periods are required to decide which treatment modality is the most effective for a given patient scenario.

Ethical approval

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee (name of institute/committee) and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed consent

Informed consent was obtained from all individual participants included in the study.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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